

Population Genetics Drill Problems

Answer Key

(1) Allele and Genotype Frequency Calculations

1.

a. $p = \frac{2 \times 245 + 210}{2 \times 500} = \frac{490 + 210}{1000} = \frac{700}{1000} = 0.70$

b. $q = 1 - p = 0.30$ (or $\frac{2 \times 45 + 210}{1000} = \frac{90 + 210}{1000} = 0.30$)

2.

a. $p = 1 - q = 1 - 0.3 = 0.7$

b. $p^2 = (0.7)^2 = 0.49$

c. $2pq = 2 \times 0.7 \times 0.3 = 0.42$

d. $q^2 = (0.3)^2 = 0.09$

3. For a rare dominant disorder, affected individuals are primarily heterozygotes. Frequency of affected $\approx 2pq \approx 2q$ (since $p \approx 1$). $2q = \frac{1}{10,000} = 0.0001$, so $q = 0.00005$.

4.

a. $q^2 = \frac{320}{2000} = 0.16$, so $q = \sqrt{0.16} = 0.4$

b. $p = 1 - q = 0.6$

c. Expected heterozygotes $= 2pq \times N = 2 \times 0.6 \times 0.4 \times 2000 = 0.48 \times 2000 = 960$

5. For X-linked loci, males express the recessive allele at frequency equal to the allele frequency.

a. Affected males $= q = 0.15$

b. Affected females $= q^2 = (0.15)^2 = 0.0225$

c. Carrier females $= 2pq = 2 \times 0.85 \times 0.15 = 0.255$

(2) Hardy-Weinberg Equilibrium Testing

6.

$$\text{a. } p = \frac{2 \times 50 + 20}{200} = \frac{100 + 20}{200} = \frac{120}{200} = 0.6, q = 0.4$$

$$\text{b. Expected: } AA = p^2 \times 100 = 0.36 \times 100 = 36, Aa = 2pq \times 100 = 0.48 \times 100 = 48, aa = q^2 \times 100 = 0.16 \times 100 = 16$$

$$\text{c. } \chi^2 = \frac{(50 - 36)^2}{36} + \frac{(20 - 48)^2}{48} + \frac{(30 - 16)^2}{16} = \frac{196}{36} + \frac{784}{48} + \frac{196}{16} = 5.44 + 16.33 + 12.25 = 34.02$$

34.02 > 3.84, therefore the population is NOT in Hardy-Weinberg equilibrium.

7. Expected heterozygosity = $2pq = 2 \times 0.6 \times 0.4 = 0.48$. Observed heterozygosity = 0.42, which is less than expected. Therefore, there is a *deficiency* of heterozygotes.

$$\text{8. } p = \frac{2 \times 640 + 320}{2000} = \frac{1280 + 320}{2000} = \frac{1600}{2000} = 0.8, q = 0.2$$

$$\text{Expected: } AA = p^2 \times 1000 = 0.64 \times 1000 = 640, Aa = 2pq \times 1000 = 0.32 \times 1000 = 320, aa = q^2 \times 1000 = 0.04 \times 1000 = 40$$

Observed counts exactly match expected, so the population is in HWE.

(3) Selection Models

$$\text{9. Mean fitness: } \bar{w} = p^2 w_{AA} + 2pq w_{Aa} + q^2 w_{aa} = (0.8)^2(1.0) + 2(0.8)(0.2)(1.0) + (0.2)^2(0.4)$$

$$\bar{w} = 0.64 + 0.32 + 0.016 = 0.976$$

$$q' = \frac{q^2 w_{aa} + pq w_{Aa}}{\bar{w}} = \frac{0.04 \times 0.4 + 0.16 \times 1.0}{0.976} = \frac{0.016 + 0.16}{0.976} = \frac{0.176}{0.976} = 0.180$$

$$\text{10. } \hat{p} = \frac{w_{Aa} - w_{aa}}{(w_{Aa} - w_{AA}) + (w_{Aa} - w_{aa})} = \frac{1.0 - 0.4}{(1.0 - 0.7) + (1.0 - 0.4)} = \frac{0.6}{0.3 + 0.6} = \frac{0.6}{0.9} = \frac{2}{3} \approx 0.667$$

$$\text{11. } \hat{p} = \frac{w_{Aa} - w_{aa}}{2w_{Aa} - w_{AA} - w_{aa}} = \frac{0.6 - 1.0}{2(0.6) - 1.0 - 1.0} = \frac{-0.4}{1.2 - 2.0} = \frac{-0.4}{-0.8} = 0.5$$

$$\text{12. } \bar{w} = p^2(1.0) + 2pq(1.0) + q^2(0) = (0.8)^2 + 2(0.8)(0.2) = 0.64 + 0.32 = 0.96$$

$$p' = \frac{p^2 + pq}{\bar{w}} = \frac{0.64 + 0.16}{0.96} = \frac{0.80}{0.96} = 0.833$$

(4) Mutation Models

$$\text{13. } \hat{q} = \frac{\mu}{\mu + \nu} = \frac{2 \times 10^{-5}}{2 \times 10^{-5} + 1 \times 10^{-6}} = \frac{2 \times 10^{-5}}{2.1 \times 10^{-5}} = \frac{2}{2.1} \approx 0.952$$

$$\text{14. } \hat{q} = \sqrt{\frac{\mu}{s}} = \sqrt{\frac{5 \times 10^{-6}}{0.3}} = \sqrt{1.667 \times 10^{-5}} \approx 0.00408$$

$$\text{15. } \hat{q} = \frac{\mu}{s} = \frac{1 \times 10^{-5}}{0.2} = 5 \times 10^{-5}$$

$$\text{16. } \hat{q} = \sqrt{\frac{\mu}{s}} = \sqrt{\frac{3 \times 10^{-5}}{0.5}} = \sqrt{6 \times 10^{-5}} \approx 0.00775$$

(5) Migration Models (Continent-Island)

17. $p'_{island} = (1-m)p_{island} + mp_{continent} = (0.92 \times 0.6) + (0.08 \times 0.9) = 0.552 + 0.072 = 0.624$

18. $p'_{island} = (1 - 0.05) \times 0.3 + 0.05 \times 1.0 = 0.95 \times 0.3 + 0.05 = 0.285 + 0.05 = 0.335$

19. $p'_{island} = (1 - 0.10) \times 0.4 + 0.10 \times 0.7 = 0.9 \times 0.4 + 0.07 = 0.36 + 0.07 = 0.43$

20. $0.52 = (1 - m) \times 0.45 + m \times 0.95$

$$0.52 = 0.45 - 0.45m + 0.95m$$

$$0.07 = 0.5m$$

$$m = \frac{0.07}{0.5} = 0.14$$